

Disproof of TLMC Conjecture 00000000578

Divisibility of characteristic polynomials of binomial matroids

Conjecture 1 (00000000578). *Let $\text{bin}(n, i)$ denote the binary matroid whose ground set is the set of i -subsets of $\{1, \dots, n\}$, represented over $\text{GF}(2)$ by the weight- i 0/1-vectors, and let $\chi_{n,i}$ denote its characteristic polynomial. For all $2 \leq i \leq n - 2$ and all integers $m \geq 1$, the value $\chi_{n,i}(-m)$ is divisible by $i!(n-i)!$, and the quotient is a nonnegative combination of Eulerian numbers.*

Theorem 2 (Verdict: FALSE). *The conjecture fails already at the first admissible parameters beyond the smallest cases. Both counterexamples below lie strictly inside the quantifier range $2 \leq i \leq n - 2$, $m \geq 1$.*

1 Attack

The characteristic polynomial of a matroid M with rank function r on ground set E is

$$\chi_M(t) = \sum_{X \subseteq E} (-1)^{|X|} t^{r(E) - r(X)}.$$

We computed $\chi_{n,i}$ from this definition directly, enumerating all subsets of the ground set and evaluating the $\text{GF}(2)$ rank by Gaussian elimination (cross-checked against brute-force span enumeration).

Counterexample 1: $\text{bin}(5, 3)$, $m = 1$. Here $r = 5$, $|E| = 10$, and

$$\chi_{5,3}(t) = t^5 - 10t^4 + 45t^3 - 105t^2 + 120t - 51.$$

At $m = 1$ the modulus is $3!2! = 12$ and

$$\chi_{5,3}(-1) = -332 = -12 \cdot 28 + 4, \quad 12 \nmid -332.$$

Counterexample 2: $\text{bin}(6, 2)$, $m = 2$. Here $r = 5$, $|E| = 15$, and

$$\chi_{6,2}(t) = t^5 - 15t^4 + 85t^3 - 225t^2 + 274t - 120.$$

At $m = 2$ the modulus is $2!4! = 48$ and

$$\chi_{6,2}(-2) = -2520 = -48 \cdot 53 + 24, \quad 48 \nmid -2520.$$

Note that $\text{bin}(6, 2)$ *does* satisfy the divisibility claim at $m = 1$, where $\chi_{6,2}(-1) = -720 = 15 \cdot 48$: the failure is specific to $m \geq 2$.

Since divisibility fails, the claimed Eulerian-quotient consequence is never reached.

2 Boundary data

Divisibility of $\chi_{n,i}(-m)$ by $i!(n-i)!$ for small admissible (n, i) and $m \in \{1, 2, 3\}$:

	$m = 1$	$m = 2$	$m = 3$
$\text{bin}(4, 2), \text{mod } 4$	OK	OK	OK
$\text{bin}(5, 2), \text{mod } 12$	OK	OK	OK
$\text{bin}(5, 3), \text{mod } 12$	FAIL (-332)	FAIL (-1263)	OK (-3624)
$\text{bin}(6, 2), \text{mod } 48$	OK (-720)	FAIL (-2520)	OK (-6720)

The smallest counterexample is $\text{bin}(5, 3)$ at $m = 1$.

3 Formal verification

The accompanying Lean 4 development (`lean4/`, core Lean only, toolchain `leanprover/lean4:v4.33.1`, no Mathlib) implements the $\text{GF}(2)$ rank of bitmask vectors and evaluates the characteristic polynomial by enumerating *all* subsets of the ground set inside Lean, so the attack numbers are computed by the kernel from the matroid data rather than assumed:

$$\text{rank}(E_{53}) = 5, \chi_{E_{53}}(-1) = -332; \quad \text{rank}(E_{62}) = 5, \chi_{E_{62}}(-2) = -2520,$$

together with the divisibility failures expressed through remainders (for positive a , $a \mid b \iff b \bmod a = 0$):

$$(-332) \bmod 12 = 4 \neq 0, \quad (-2520) \bmod 48 = 24 \neq 0, \quad 332 \bmod 12 = 8, \quad 2520 \bmod 48 = 24.$$

The boundary values $\chi_{5,3}(-2) = -1263$, $\chi_{5,3}(-3) = -3624$, $\chi_{6,2}(-1) = -720$, $\chi_{6,2}(-3) = -6720$ are verified as well. An axiom audit (`Check.lean`) confirms that every definition and theorem *does not depend on any axioms* and the development contains no `sorry`.

4 Reproducing

```
python3 reproduce.py          # independent recomputation + assertions
cd lean4 && lake build         # kernel-computed attack numbers
lake env lean Check.lean      # zero-axiom audit
```